Lab - III
Srn: PES2UG24CS311
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1:

```
#include<stdio.h>
int main(){
	printf("Enter a number: \n");
	int num;
	scanf("%d", &num);
	int sum = 0, temp = num, x;
	int digit = 0;
	while(temp){
		digit++;
		temp /=10;
	}
	temp = num;
	while(num){
		x = num % 10;
		sum += pow(x, digit);
		num /= 10;
	}
	if(sum == temp)
		printf("It is armstrong number\n");
	else
		printf("It is not an armstrong\n");
	return 0;
}
```

2:

```
#include<stdio.h>
int main(){
	printf("Enter a number: \n");
	int num;
	scanf("%d", &num);
	int sum = 0, temp = num, x;
	temp = num;
	int digit =0;
	for(int i = 1; i <= num; i++){
		temp = i;
		while(temp){
		digit++;
		temp /=10;
		}

		digit--;
		temp = i;
		//printf("digit: %d\n", digit);
		//printf("temp: %d i: %d\n", temp, i);
		while(temp){
			sum += (temp % 10) * pow(10, digit);
			//printf("sum: %d\n", sum);
```

```
                    temp /= 10;
                    digit--;
            }
            if(sum == i)
                    printf("%d is palindrome\n", i);
            digit = 0;
            sum = 0;
    }
    return 0;
}
```

3:

```
#include<stdio.h>
int main(){
    printf("Enter a number: \n");
    int num1, num2,n;
    scanf("%d", &n);
    int is_prime_1 = 1;
    int is_prime_2 = 1;
    int is_prime_3 = 1;
    int num3;
    for(int j = 2; j <= n; j++)
    {
            num1 = j;
            for(int l = 2; l<=n; l++)
            {
                    num2 = l;
                    is_prime_1 = 1;
                    is_prime_2 = 1;
                    is_prime_3 = 1;
                    //printf("for (%d, %d):\n", num1, num2);
                    for(int i = 2; i < (num1); i++){
                            if (num1 % i == 0){
                                    is_prime_1 = 0;
                                    //printf("%d is not prime\n", num1);
                                    }
                    }
                    for(int i = 2; i < (num2); i++){
                            if (num2 % i == 0){
                                    is_prime_2 = 0;
                                    //printf("%d is not prime\n", num1);
                                    }
                    }
                    if (is_prime_1 == 1 && is_prime_2 == 1){
                            printf("The numbers (%d, %d) are both prime and:\n", num1,
num2);
                            num3 = num2 - num1;
                            if(num3<0)
                                    num3 = -1 * num3;
                            for(int i = 2; i < (num3); i++){
                            if (num3 % i == 0)
                                    is_prime_3 = 0;
```

```
                }
                if(num3 == 1 || num3 == 0)
                        is_prime_3 = 0;
                if (is_prime_3 == 1){
                        printf("They are super primes\n");
                        }
                else
                        printf("They are Not super Primes\n");
                printf("\n\n");

            }
        }
    }


    return 0;
}
```

4:

```
#include<stdio.h>
int main(){
    printf("Enter 0 to stop the program: \n");
    int num = 0;
    int factor_sum;
    while(1){
        scanf("%d", &num);
        if(num == 0)
            break;
        factor_sum = 0;
        for(int i = 1; i <= (num/2); i++){
            if (num % i == 0)
                factor_sum += i;
        }
        if(factor_sum == num)
            printf("%d is a perfect number\n", num);
        else
            printf("%d is not a perfect number\n", num);
    }
    return 0;
}
```